

Aaron Heumphreus

Project 1

10/6/2024

CS 5130

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File Edit Selection View Go Run Terminal Help
Project_1_5130

EXPLORER
OPEN EDITORS
sort_test.py
PROJECT_1_5130
.venv
  Include
  Lib
  Scripts
    activate
    activate.bat
    activate.ps1
    deactivate.bat
    pip.exe
    pip3.12.exe
    pip3.exe
    python.exe
    pythonw.exe
    pyvenv.cfg
  sort_test
    Activate.ps1
    Include
    Lib
    Scripts
    pyvenv.cfg
    python.txt
    README.md
    sort_test.py

sort_test.py
54 #list for containing insert sort times with range 10000x1
55 insert_sort_times: list[int]=[]
56 #list for containing bubble sort times with range 10000x1
57 bubble_sort_times: list[int]=[]
58
59
60 #Loop for bubble_sort_times population
61 for j in range(LIST_NUM):
62     #Start Time
63     start: int=time.perf_counter_ns()
64     #generates 10,000 integers between 1 and 1000
65     input: list[int] = [random.randrange(1, 1000, 1) for i in range((j+1)*10000)]
66     #Calls bubble sort using input
67     bubble_sort(input)
68     #Stop Time
69     stop: int=time.perf_counter_ns()
70     #Appends difference between start and stop times to bubble_sort_times
71     bubble_sort_times.append(stop-start)
72
73 #Loop for insert_sort_times population
74 for j in range(LIST_NUM):
75     #Start Time
76     start: int=time.perf_counter_ns()
77     #generates 10,000 integers between 1 and 1000
78     input: list[int]= [random.randrange(1, 1000, 1) for i in range((j+1)*10000)]
79     #Calls insert sort using input
80     insert_sort(input)
81     #Stop Time
82     stop: int=time.perf_counter_ns()
83     #Appends difference between start and stop times to bubble_sort_times
84     insert_sort_times.append(stop-start)
85
86 print("Insert Sort Times per 10,000", insert_sort_times)
87 print("Bubble Sort Times per 10,000", bubble_sort_times)

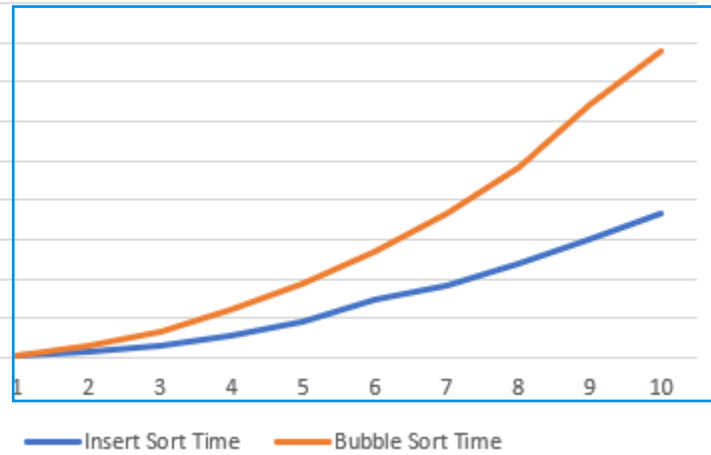
TERMINAL
File "C:\Users\Aaron\Documents\Project1_5130\Project_1_5130\sort_test.py", line 65, in <module>
TypeError: type 'int' is not subscriptable
(sort_test) PS C:\Users\Aaron\Documents\Project1_5130\Project_1_5130> py sort_test.py
[3763803700, 14660682800, 1779277500, 7162064300]
130> py sort_test.py
Insert Sort Times per 10,000 [1799591000]
Bubble Sort Times per 10,000 [3738563100]
(sort_test) PS C:\Users\Aaron\Documents\Project1_5130\Project_1_5130> py sort_test.py
Insert Sort Times per 10,000 [1822236900, 7185284600, 16216831400, 29877352400, 45615104300, 73672928400, 90477733400, 118555337400, 149944221600, 182501632300]
Bubble Sort Times per 10,000 [3645107900, 14476303500, 33100890600, 59956704000, 92971393400, 135605328100, 184170072000, 241489963400, 321478346900, 388021300500]
(sort_test) PS C:\Users\Aaron\Documents\Project1_5130\Project_1_5130>

Ln 78, Col 85 Spaces: 4 UTF-8 CRLF () Python 3.12.7 (.venv\venv)
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Insert Sort Time	Bubble Sort Time
1,822,236,900	3,645,107,900
7,185,284,600	14,476,303,500
16,216,831,400	33,100,890,600
29,077,352,400	59,956,704,000
45,615,104,300	92,971,393,400
73,627,978,400	135,605,328,100
90,477,733,400	184,170,072,000
118,555,337,400	241,489,963,400
149,944,223,600	321,478,346,900
182,501,632,300	388,021,300,500

450,000,000,000
400,000,000,000
350,000,000,000
300,000,000,000
250,000,000,000
200,000,000,000
150,000,000,000
100,000,000,000
50,000,000,000
-

Sort Time per 10,000 entries



```

if __name__ == "__main__":
    import random
    import time

def bubble_sort(sort_list: list[int]):
    """This function sorts out a list of ints in place using Bubble Sort Format.
    -----
    Params
    -----
        sort_list: list[int]
            list of ints
    -----
    Returns
    -----
        none:
            sorts list in place
    """
    #increments through list via i
    for i in range(len(sort_list)-1):
        #increments through list j of i
        for j in range(i+1,len(sort_list)):
            #conditional to chheck if position j is less than j-1, then swaps if true
            if sort_list[j] < sort_list[j-1]:
                #tmp for temp storage for int at value j
                tmp:int=sort_list[j]
                sort_list[j]=sort_list[j-1]
                sort_list[j-1]=tmp

def insert_sort(sort_list: list[int]):

```

```

"""This function sorts out a list of ints in place using Insert Sort Format.
-----
Params
-----
    sort_list: list[int]
        list of ints
-----
Returns
-----
    none:
        sorts list in place.
"""
for j in range(2,len(sort_list)):
    key=sort_list[j]
    i=j-1
    while i > 0 and sort_list[i] > key:
        sort_list[i+1]=sort_list[i]
        i=i-1
    sort_list[i+1]=key

#Constant used for determining the number of times each input is run
LIST_NUM: int=10
#list for containing insert sort times with range 10000xi
insert_sort_times: list[int]=[]
#list for containing bubble sort times with range 10000xi
bubble_sort_times: list[int]=[]

#Loop for bubble_sort_times population
for j in range(LIST_NUM):
    #Start Time
    start: int=time.perf_counter_ns()

```

```
#generates 10,000 integers between 1 and 1000
input: list[int] = [random.randrange(1, 1000, 1) for i in range((j+1)*10000)]
#Calls bubble_sort using input
bubble_sort(input)
#Stop Time
stop: int=time.perf_counter_ns()
#Appends difference between start and stop times to bubble_sort_times
bubble_sort_times.append(stop-start)

#Loop for insert_sort_times population
for j in range(LIST_NUM):
    #Start Time
    start: int=time.perf_counter_ns()
    #generates 10,000 integers between 1 and 1000
    input: list[int]= [random.randrange(1, 1000, 1) for i in range((j+1)*10000)]
    #Calls insert_sort using input
    insert_sort(input)
    #Stop Time
    stop: int=time.perf_counter_ns()
    #Appends difference between start and stop times to bubble_sort_times
    insert_sort_times.append(stop-start)

print("Insert Sort Times per 10,000", insert_sort_times)
print("Bubble Sort Times per 10,000", bubble_sort_times)
```